

ECS 119: Data Processing Pipelines

Fall 2026

Welcome! (Am I in the right place?)

- This is a **required course** for data science majors
- Open as a CS non-major elective (100-119) and to MS students
- Cap: 90, 0 currently waitlisted
- 4 units

Calendar Items

- **Lectures:** MWF 3:10-4pm in Wellman Hall 126
- **Discussion section:** Wednesdays at 11am in Olson Hall 206 (from next week)
- **Office hours:** Fridays after class
- **Final exam:** 12/11/2026 10:30am

Syllabus + schedule + all of the above information is on [the class repository](#)

Please join [ECS 119 on Piazza](#) (I will remind you again later)

About the Instructor



Prof. Stanford
Started at Davis July 2023



[Website](#)

Class TA



[Muhammad Hassnain](#)

Discussion section: Wednesdays at 11am i
Olson Hall 206

Plan for today

1. Course introduction
2. FAQ, Syllabus and logistics
3. Questions for me

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What is this class about?

ECS 119: Data Processing Pipelines

[Home](#) • [Schedules and Classes](#) • [ECS 119: Data Processing Pipelines](#)

Subject

ECS 119

Title

Data Processing Pipelines

Status

Active

Units

4.0



Discussion Question

You have compiled a dataset of **water level information** for various **lakes, bays, reservoirs, and oceans** in California and across the United States. Each **row** in the **dataset of readings** contains measured depth, temperature, humidity, and saline level. New data comes in weekly from various sites. You want to put together an app which identifies sites that are at-risk and in danger or water level decline.

1. Which of the following **languages and tools (of those you recognize)** would you like to use to build your app and why? Python, R, Excel, SQL, C++, Rust, Other
2. Do you need access to any **subscriptions (of those you recognize)** to build your app? (AWS, Google Cloud, website/server costs, AI credits)
3. Suppose the data consisted of **10K (10,000) readings, 10M (10,000,000) readings, or 10B (10,000,000,000) readings**. How would your answers to the above change in each of these scenarios?

Let's discuss first; we will share our answers on the following slide.

Sharing Your Answers

forms.gle/pYhTyDu8YuYTmwtB9

In-class polls are for completion credit only.



A common theme

Software components, design constraints, tools and development techniques differ depending on the size of the dataset...

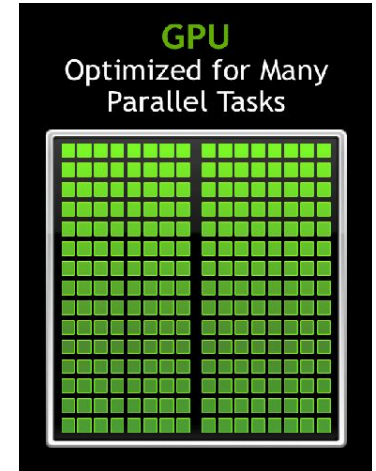
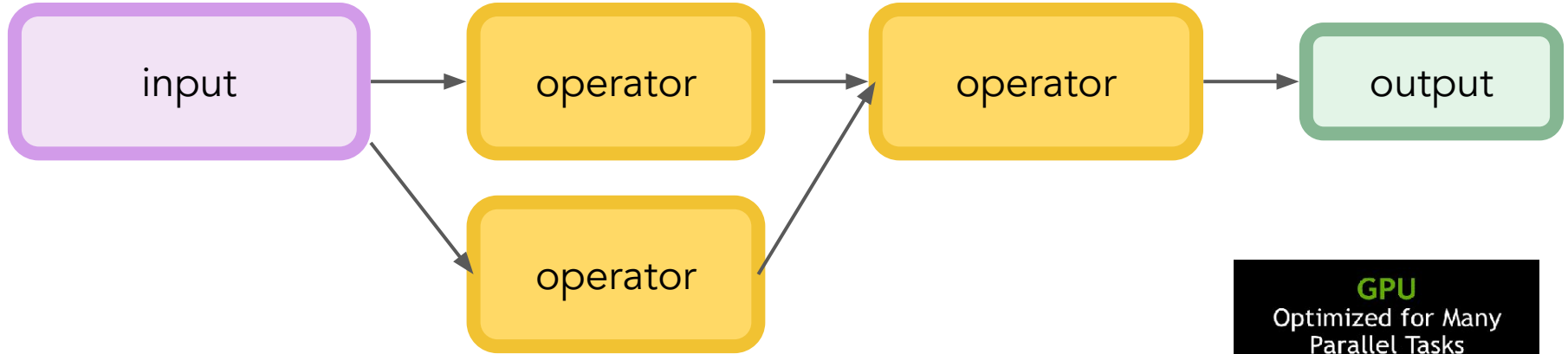
- ... 10K datapoints
- ... 10M datapoints
- ... 10B datapoints



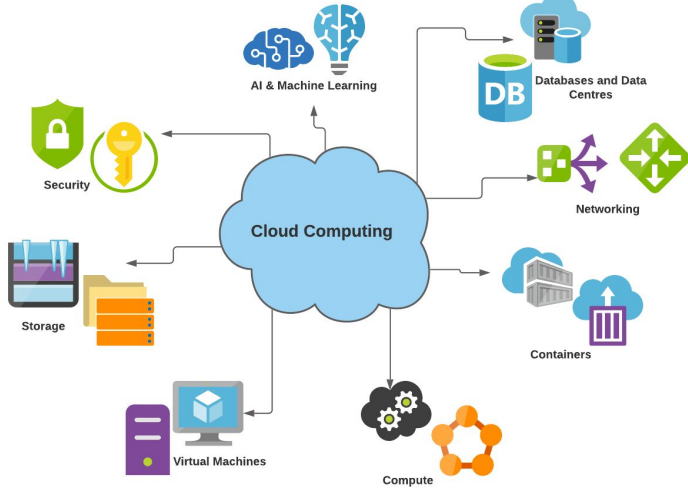
Building applications for large datasets?



Parallel computing



Distributed computing



How did hackers compromise my EC2 instance?

Asked 8 years, 6 months ago Modified 2 years, 8 months ago Viewed 8k times

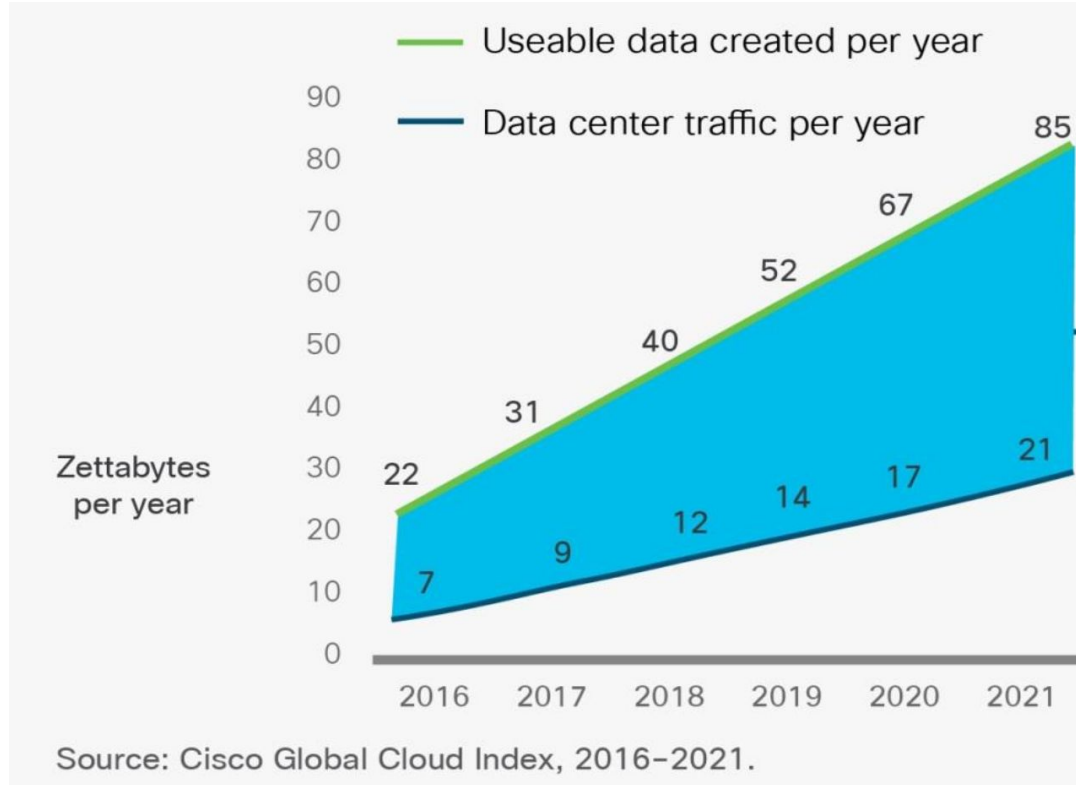


11



My EC2 instance was hacked recently. It doesn't really matter as I'm just starting my website and there was no sensitive information on my server yet, but I do plan for there to be in the future. I am going to terminate the compromised server and set up another one in order to secure my website.

Real-time computing



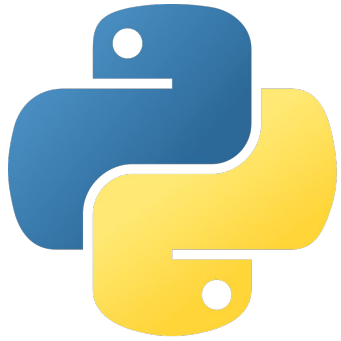
More data
produced
than can be
feasibly
stored

What this course is about

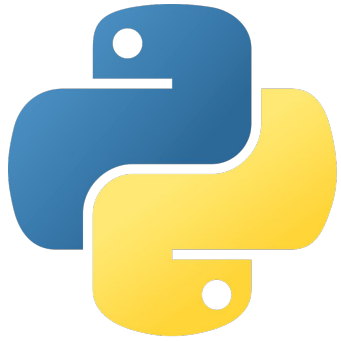
Principles and practice of working with **large datasets**...



... using **real tools** used in industry



... using **real tools** used in industry



✨ Hands-on Approach ✨



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FAQ

Q: What are the prerequisites?

ECS 116 or 165A

Ability to program [FizzBuzz](#)

Q: Is this an AI/ML course?

Short answer: No

Long answer: No, but the tools and software used in this class are of fundamental importance for building, training, and deploying large ML models in industry

(It's more related to something called “ML Engineering” - the tools in this class can be for using/deploying models on large datasets)

Q: Can I take this class as an elective?

Answer: **Yes**

Disclaimer: Non-major CS class – **does not count as an upper-division elective***

- * I am hoping to get this changed, however, this probably won't happen for this quarter

Should I? **The material will be especially relevant to you if:**

- You want to know about the fundamental principles & practice behind large-scale data processing tasks
- You want to gain hands-on experience with the tools used in industry

Q: Can you add me off the waitlist / waive my prerequisites?

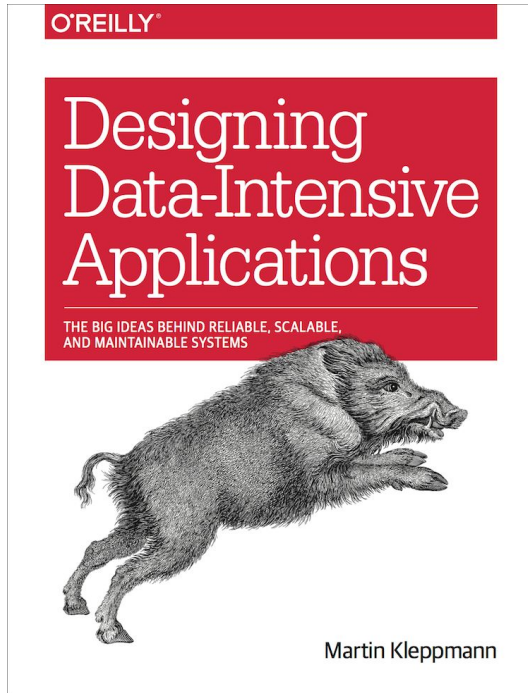
Unfortunately I am not allowed to! - all requests have to go through undergraduate advising

csugadvising@ucdavis.edu

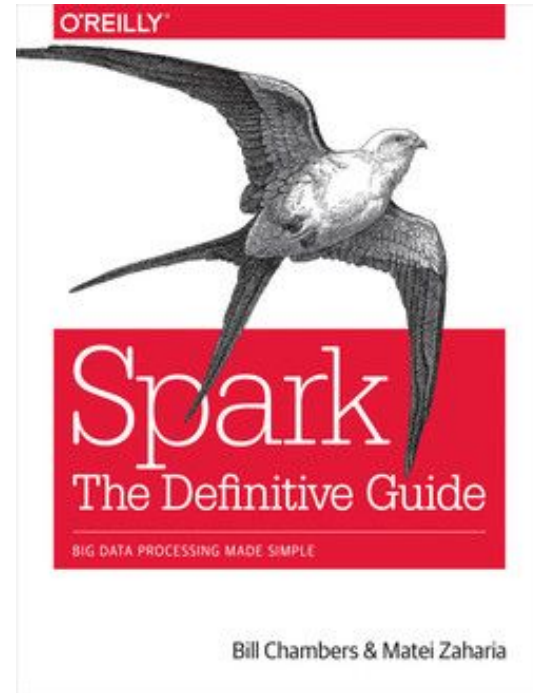
For the waitlist (if applicable) - I am sorry I can't guarantee a spot. If you are in the first 5-10 spots and you attend consistently there is a good chance that you could get in.

Syllabus and logistics

Textbooks (Optional) – *Newly updated*



DDIA



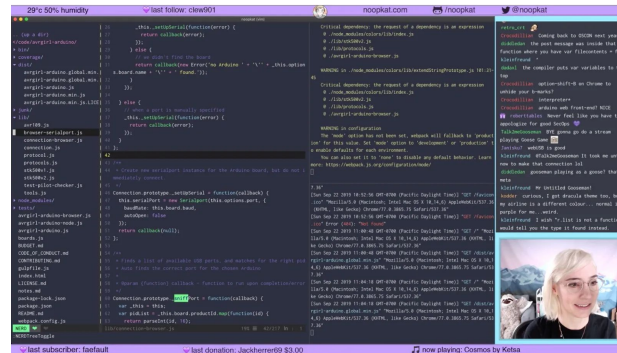
Spark Book

Lectures

- I generally lecture with live coding (not slides)
- Working with real tools can only be learned by **doing :-)**

Attendance: Lectures and discussion section are encouraged via in-class polls (see later)

- Not required
- In-class polls can be made up at any time



To follow along...

<https://github.com/DavisPL-Teaching/119>

```
> git clone git@github.com:DavisPL-Teaching/119.git
```

```
> git pull
```

Piazza

Please join the [Piazza](#)

Grade breakdown

- **Participation (10%)**: via in-class quizzes
- **Homeworks (25%)**: about 2-3 programming assignments, plus homework 0
- **Midterm (25%)**: covering the first half and main concepts of the course
- **Final Exam (40%)**: covering all topics covered in the course.

Minimum grade cutoffs: 93%=guaranteed A, 90%=guaranteed A-, etc.

Exams may be curved - I reserve the right to lower the maximum score

Attendance and participation (10%)

Fill out the in-class polls (participation points only)

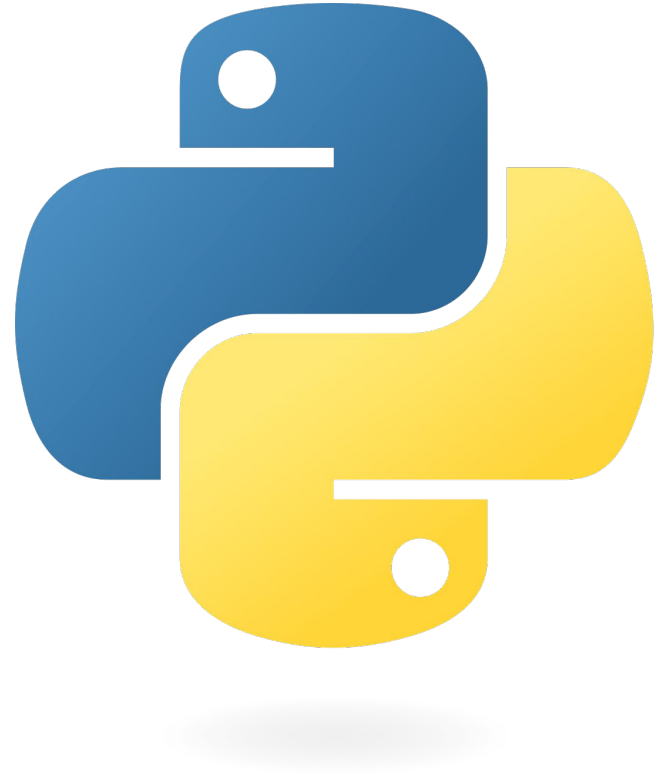
If you are sick: Starting from the first day of class, you may join the class remotely via Zoom (the quality may not be as good)

If you miss class: Lectures are recorded. You can make up the in-class polls at any time



Homeworks (25%)

We will be working in Python for most of the course

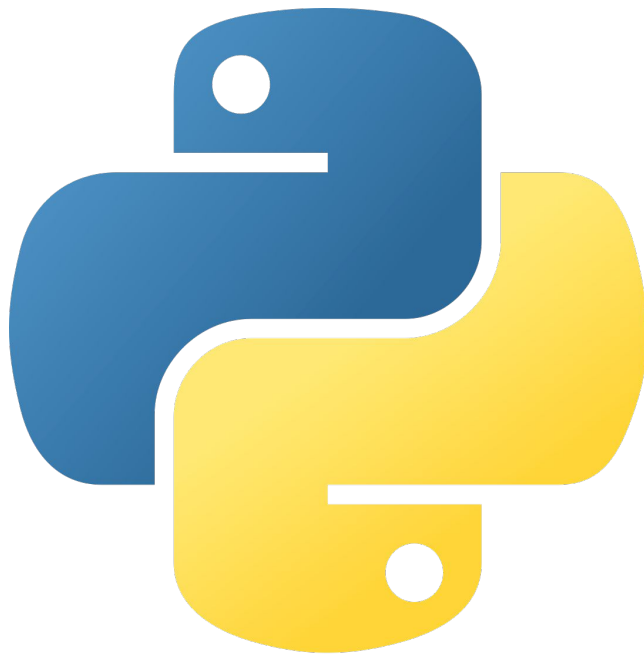


Homeworks (25%)

We will be working in Python for most of the course

Planned HWs:

- HW1 (Pandas): Data wrangling, data cleaning, and data validation
- HW2 (PySpark): Parallel processing
- Possibly one more



Homework 0: installation help



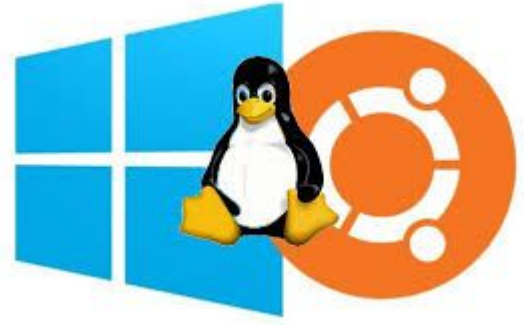
(submission link posted soon, probably Friday)



Homework 0: installation help

I recommend using **MacOS or Linux**

If you are on Windows, I recommend



Windows Subsystem for Linux (WSL)

- Well-engineered tool and very useful to know about when doing software development work in the real world!
- <https://learn.microsoft.com/en-us/windows/wsl/install>

Homework grading

Most important: please run your code

- Software engineering is about running software!
- Experienced engineers will run and test their code frequently
- We can't give points to code that does not run 😞

Exams (midterm: 25%, final: 40%)

- Closed book, one-sided cheat sheet allowed
- Practice questions given! Come to office hours! The goal is for you to pass.

Planned midterm date: Friday, November 6

Final: Friday, December 11

Platforms

Class discussion, Q+A, and announcements:

- [ECS 119 on Piazza](#)
 - **Don't email me, post to Piazza!**
 - Make your post public and (if you prefer) anonymous

Homeworks and Exams: Gradescope

AI Policy

AI is an increasingly powerful tool: I allow it, as I think understanding it is important for your future.

Please use it carefully: Many students have found that they learn better when limiting or restricting their use of AI early on.

Don't use AI to copy and paste entire homework solutions.

The midterm and final exam will be in-class and closed-book.

Collaboration Policy

- **Collaboration is encouraged!**
- Groups of size up to 4
- Everyone should submit their own solution
- Please list your collaborators at the top of your homework

Scheduling note

Please note: there will be **no class on Friday, October 9** as I will be away at a conference.

Outline of planned lectures

- Lecture 1: Introduction to data processing pipelines, input data sources
- Lecture 2: The Shell, Git, software engineering tools
- Lecture 3 (Optional): Testing and Validation, some pandas
- Lecture 4: Parallelism and concurrency
- Lecture 5: Distributed pipelines
- Lecture 6: Real-time and streaming pipelines
- Lecture 7: Cloud computing

[View an example schedule \(from a previous year\)](#)

Communication reminders

TA: **Hassnain**

My office hours: Fridays after class (415-5pm, see Piazza)

TA office hours: TBD on Piazza

Please use Piazza for questions (if you email or send me a Canvas message, I will tell you to post to Piazza :-)

Respect and discrimination

✨ **Please be nice!** ✨

Include everyone in group discussions

Reach out to me in case of any problems

Late policy

Polls and HW0: can be made up

- You don't need to email me about this, just submit the poll! :)

For HW, I cannot guarantee that late work is graded. BUT: I encourage you to keep working on an assignment if you didn't finish it by the deadline. Gradescope will allow you to submit a few days late, and we may choose to grade these late submissions, at our discretion.

Some homeworks may be long - please start early.

Plan ahead! Ask for an extension if you need it, ahead of time.

Other reminders

Job scams:

- [Job Scams | Career Center](#)
- [UC Davis Phish Bowl](#)
- [Policies against harassment and discrimination](#)
- [List of resources for student health and well-being](#)

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Questions for me?

Reminder

Please join the [Piazza](#)

Learning objectives

- Understand the components, techniques, tools, and performance metrics of setting up data processing jobs in Python and related tools
- Understand the concepts of parallelism, types of parallelism, and parallelization mechanisms, via tools like MapReduce and Spark.
- Understand how software engineering tools and configuration are integrated into a data project, via tools like Git and the shell
- Understand the concepts of distributed computing and distributed data processing, including distributed consistency requirements, and how it manifests in real-world applications.
- Understand advanced topics including programming over real-time and streaming data sources and using cloud platforms such as AWS, Azure, and Google Cloud.